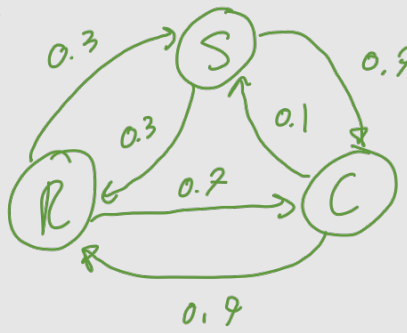
Resolver



$\Rightarrow$

R	S	C
$P_{RS}=0.3$	$P_{SR}=0.3$	$P_{CS}=0.1$
$P_{RC}=0.7$	$P_{SC}=0.7$	$P_{CR}=0.9$

Podemos decir que

$\pi_R = \pi_S P_{SR} + \pi_C P_{CR}$

$\pi_S = \pi_R P_{RS} + \pi_C P_{CS}$

$\pi_C = \pi_R P_{RC} + \pi_S P_{SC}$

$1 = \pi_R + \pi_S + \pi_C$

$\rightarrow$

0) $\pi P - \pi = 0$

$$\begin{array}{c} \\ a. \\ b. \\ c. \\ \hline d. \end{array} \begin{array}{c} \begin{array}{ccc} \pi_R & \pi_S & \pi_C \end{array} \\ \begin{pmatrix} -1 & P_{SR} & P_{CR} \\ P_{RS} & -1 & P_{CS} \\ P_{RC} & P_{SC} & -1 \\ \hline 1 & 1 & 1 \end{pmatrix} \end{array} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ \hline 1 \end{pmatrix}$$

Luego

$a. = a. - d.\left(\frac{P_{CR} + P_{CS} P_{SR}}{1 + P_{CS}}\right)$

$c. = c. - d.\left(\frac{P_{RC} + P_{RS} \cdot P_{SC}}{1 + P_{RS}}\right)$

Sea $a = a. + b.(P_{SR})$ $\quad b. = b.$

$c. = c. + b.(P_{SC})$

$d. = d. + b.$

$$\begin{pmatrix} P_{RS} \cdot P_{SR} - 1 & 0 & P_{CR} + P_{CS} P_{SR} \\ P_{RS} & -1 & P_{CS} \\ P_{RC} + P_{RS} \cdot P_{SC} & 0 & P_{CS} \cdot P_{SC} - 1 \\ 1 + P_{RS} & 0 & 1 + P_{CS} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} P_{RS} \cdot P_{SR} - 1 & 0 & 0 \\ -(1+P_{RS})\left(\frac{P_{CR} + P_{CS} \cdot P_{SR}}{1 + P_{CS}}\right) & -1 & P_{CS} \\ P_{RS} & & \\ & 0 & \frac{P_{CS} \cdot P_{SC} - 1}{-(1+P_{CS})\left(\frac{P_{RC} + P_{RS} \cdot P_{SC}}{1 + P_{RS}}\right)} \\ 0 & 0 & \end{pmatrix} = \begin{pmatrix} -\frac{P_{CR} + P_{CS} P_{SR}}{1 + P_{CS}} \\ 0 \\ -\frac{P_{RC} + P_{RS} \cdot P_{SC}}{1 + P_{RS}} \end{pmatrix}$$

$$\pi_R = \frac{\left(\frac{-P_{CR} + P_{CS} \cdot P_{SR}}{1 + P_{CS}}\right)}{P_{RS} \cdot P_{SR} - 1 - (1 + P_{RS})\left(\frac{P_{CR} + P_{CS} \cdot P_{SR}}{1 + P_{CS}}\right)}$$

$\leftarrow PR_N$ $\quad$ $\hookrightarrow PR_D$

$$\pi_C = \frac{-\left(\frac{P_{RC} + P_{RS} \cdot P_{SC}}{1 + P_{RS}}\right)}{P_{CS} \cdot P_{SC} - 1 - (1 + P_{CS})\left(\frac{P_{RC} + P_{RS} \cdot P_{SC}}{1 + P_{RS}}\right)}$$

$\leftarrow PC_N$ $\quad$ $\hookrightarrow PC_D$

$$\pi_S = 1 - (\pi_R + \pi_C)$$

[2] ✓ 0s

```
Prs = 0.3
Prc = 0.7
Psr = 0.3
Psc = 0.7
Pcr = 0.9
Pcs = 0.1
```

← Probabilidades

[11] ✓ 0s

```
PR_N = -(Pcr + Pcs*Psr)/(1+Pcs)
print(PR_N)
PR_D = Prs*Psr -1 -(1+Prs)*( (Pcr+Pcs*Psr)/(1+Pcs) )
print(PR_D)
PR = PR_N/PR_D
print("PR = ",PR)
```

```
-0.8454545454545455
-2.0090909090909093
PR =  0.420814479638009
```

PR_N ← PR_D ← π_R

[13] ✓ 0s

```
PC_N = -(Prc+Psr*Psc)/(1+Prs)
print(PC_N)
PC_D = Pcs*Psc -1 -(1+Pcs)*( (Prc+Prs*Psc)/(1+Prs) )
print(PC_D)
PC = PC_N/PC_D
print("PC = ",PC)
```

```
-0.7
-1.7000000000000002
PC =  0.41176470588235287
```

PC_N ← PC_D ← π_C

[14] ✓ 0s

```
PS = 1-(PR+PC)
print("PS = ",PS)
```

```
PS =  0.16742081447963808
```

π_S

Imagen de los calculos realizados según las ecuaciones obtenidas.

A continuación se muestra el procedimiento con Gauss - Seidel.

Suponiendo los valores π^0

$\pi_R^0 = 0.4$ $\pi_S^0 = 0.2$ $\pi_C^0 = 0.4$

A)

1A)

- $\pi_R' = \frac{\pi_S P_{SR} + \pi_C P_{CR}}{1 - 0 \leftarrow (R,R) \not\in \therefore P_{RR} = 0} = 0.2(0.3) + 0.3(0.9) = 0.06 + 0.27 = \underline{0.42}$

- $\pi_S' = \pi_R' P_{RS} + \pi_C P_{CS} = 0.33(0.3) + 0.3(0.1) = 0.099 + 0.03 = \underline{0.166}$

- $\pi_C' = \pi_R' P_{RC} + \pi_S' P_{SC} = 0.33(0.7) + 0.129(0.7) = 0.231 + 0.0903 = \underline{0.4111}$

2A) $\pi_i = \sum_{i \in \{R,S,C\}} \pi_i \Rightarrow \underline{\pi_R = 0.4216} \quad \underline{\pi_S = 0.1666} \quad \underline{\pi_C = 0.4117}$

3A) $\varepsilon = \sum_j |\pi_j - \pi_j'| \quad \forall j \in \{R,S,C\} \Rightarrow \underline{\varepsilon = 0.0667\ldots}$

4A) $\varepsilon < 1 \times 10^{-6}$? $\Rightarrow$ 1B) $\leftarrow$ ε es muy grande.

B

1B) $\pi_R^2 = \pi_S' P_{SR} + \pi_C' P_{CR} = 0.1666(0.3) + 0.4117(0.9) = \underline{0.4205\ldots}$

$\pi_S^2 = \pi_R^2 P_{RS} + \pi_C' P_{CS} = (0.4205)(0.3) + (0.4117)(0.1) = \underline{0.1673\ldots}$

$\pi_C^2 = \pi_R^2 P_{RC} + \pi_S^2 P_{SC} = (0.4205)(0.7) + (0.1673)(0.7) = \underline{0.41176\ldots}$

2B) $\underline{\pi_R = 0.42079} \quad \underline{\pi_S = 0.1674} \quad \underline{\pi_C = 0.41176}$

3B) $\underline{\varepsilon = 0.0016\ldots}$

4B) $\varepsilon \not< 1 \times 10^{-6} \Rightarrow$ 1C)

C

1C) $\pi_C^3 = \pi_R^2 P_{RC} + \pi_S^2 P_{SC} = (0.1678)(0.7) + (0.4117)(0.7) = \underline{0.420819\ldots}$

$\pi_S^3 = \pi_R^2 P_{RS} + \pi_C^3 P_{CS} = (0.4208)(0.3) + (0.41176)(0.1) = \underline{0.1674\ldots}$

$\pi_R^3 = \pi_S^3 P_{SR} + \pi_C^3 P_{CR} = (0.4208)(0.3) + (0.1674)(0.9) = \underline{0.4117\ldots}$

2C) $\underline{\pi_R = 0.420819\ldots} \quad \underline{\pi_S = 0.16742} \quad \underline{\pi_C = 0.41176\ldots}$

3C) $\underline{\varepsilon = 0.0000337}$

4C) $\varepsilon \not< 1 \times 10^{-6} = 1D)$

C

1c) $\pi_C^4 = \pi_R^3 P_{RC} + \pi_S^3 P_{SC} = (0.1678)(0.2) + (0.81176)(0.1) = 0.420843\ldots$

$\pi_S^4 = \pi_R^4 P_{RS} + \pi_C^3 P_{CS} = (0.4208)(0.3) + (0.81176)(0.1) = 0.1674207\ldots$

$\pi_R^4 = \pi_S^4 P_{SR} + \pi_C^4 P_{CR} = (0.4208)(0.3) + (0.1674)(0.7) = 0.8117696\ldots$

2c) $\pi_R = 0.42081994\ldots$ $\pi_S = 0.1674208$ $\pi_C = 0.81176470S\ldots$

3c) $\varepsilon = 0.0000007096$

4c) $\varepsilon < 1 \times 10^{-6} = 1_D \Rightarrow$ End